W2 · Regression with Multiple Features

Multiple linear regression · Vectorization · Feature scaling · Polynomial features

Name _____ Date _____

1. Concept Check

Choose the single best answer for each question.

1. You want to predict house prices using size, room count, and house age. Which description is most appropriate?

(1) Copying the target price three times
(2) Necessarily keeping only one of the three features
(3) Removing the cost function
(4) Representing one house using multiple input features

2. In $\hat{y} = w_1 x_1 + w_2 x_2 + b$, what does each weight do?

(1) Controls how much its feature contributes to the prediction
(2) Determines the name of its feature
(3) Determines only the number of data rows
(4) Changes the actual target arbitrarily

3. If $x = (2, 3)$, $w = (4, 1)$, and $b = 1$, what is $\hat{y} = w^T x + b$?

(1) 10
(2) 11
(3) 12
(4) 14

4. What is the main reason to use vectorization?

(1) To train on different targets
(2) To express the same calculation concisely and efficiently with array operations
(3) To replace every feature with one constant
(4) To guarantee zero model error

5. House size is measured in thousands, while room count is a single digit. Why apply feature scaling before gradient descent?

(1) To make all actual house prices equal
(2) To remove features with small values
(3) To make size and room count mean the same thing
(4) To reduce differences in numerical ranges and help training progress smoothly

1. Concept Check / Continued

Choose the single best answer for each question.

6. A feature has mean 10 and standard deviation 2. What is the standardized value of 14 using $(x - \mu)/\sigma$?

(1) 2
(2) 4
(3) 7
(4) 12

7. You standardize a new house for prediction. Which mean and standard deviation should you use?

(1) Values calculated from only the new house
(2) New random values at each prediction
(3) The saved values calculated from the training data
(4) The mean and standard deviation of target prices

8. During training, the cost keeps increasing and oscillating strongly. What should you try first?

(1) Increase the learning rate further
(2) Reduce the learning rate and monitor the cost
(3) Replace all actual targets with zero
(4) Stop recording the cost

9. Price varies with house size x along a curve. Which approach from this lesson could you try?

(1) Replace every input with 1
(2) Keep every weight fixed at 0
(3) Randomly change the target for the same input
(4) Add a new feature such as x^2

10. Why can $\hat{y} = w_1 x + w_2 x^2 + b$ be treated as linear regression even though it is curved with respect to x?

(1) The parameters w_1, w_2, b enter linearly
(2) Every curve is a straight line
(3) Inputs must be 0 or 1
(4) The weights are not learned

2. Think Like an ML Scientist

Question 11 · Read the pseudocode and find why the model does not work as intended.

You predict house prices using size and room count. During training you standardized each feature and obtained a small training error. Predictions for new houses are unreasonable. Below, mu and sigma are the feature means and standard deviations, and every sigma is positive. The training function works correctly.

```
mu = column_mean(X_train)
sigma = column_std(X_train)
X_scaled = (X_train - mu) / sigma
w, b = fit_linear(X_scaled, y_train)
x_new = [1500, 3]
prediction = dot(x_new, w) + b
```

11(1) Compare the training input with the prediction input. Explain which processing step is missing.

11(2) Add one preprocessing line and correct the final line. State which data should supply the statistics.

3. Open Question

Question 12 · There is no single correct answer. Describe your method and reasoning.

A tool converts sentences into equal-length lists of numbers. You can also add sentence length and the number of exclamation marks as features. To find similar customer inquiries, which features would you use and how would you compare them?

Explain what you would do if the features have very different numerical ranges, and what you would compare before and after adding features. You may assume that you need to investigate what the numbers produced by the tool mean.

Equations and code are optional. You may answer with text, a diagram, or a simple flowchart.